

Laxshika Sivalingam

Software Engineer

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Portfolio | github.com/laxshi24 | linkedin.com/in/laxshika

Professional Summary

Software Engineering graduate with a strong foundation in full-stack web development and scalable application design. Experienced in developing MERN stack applications, responsive user interfaces, and RESTful APIs through academic research and industry projects. Proficient in Python, Java, React, Node.js, Spring Boot, Flutter, and MongoDB, with hands-on experience in software testing, UI/UX design, and Agile development practices. A quick learner with strong analytical, problem-solving, and collaboration skills, passionate about building innovative, high-quality software solutions.

Education

Sri Lanka Institute of Information Technology (SLIIT), Malabe

2022 – 2026

BSc (Hons) in Information Technology – Special

Udupiddy Girls' College, Udupiddy

2016 / 2019

G.C.E. Ordinary Level & Advanced Level

Experience

Software Engineer Intern

Mar 2025 – Sep 2025

Gamage Recruiters (PVT) LTD

- Contributed to the end-to-end development of a large-scale web application using the MERN Stack (MongoDB, Express.js, React.js, Node.js) following Agile software development practices.
- Designed UI/UX wireframes and transformed Figma designs into responsive, reusable React.js components for desktop and mobile platforms.
- Implemented complete CRUD functionality for Instructor, Exam, and Quiz Management modules with search, filtering, form validation, and PDF report generation.
- Developed and integrated RESTful APIs with MongoDB, enabling efficient frontend-backend communication and data management.
- Performed API testing with Postman and resolved frontend, backend, database, and responsive UI issues through debugging and optimization.
- Applied Git and GitHub workflows, including branching, pull requests, code reviews, merge conflict resolution, and collaborative development.
- Participated in requirement analysis, sprint planning, daily stand-ups, code reviews, and technical discussions while managing tasks through ClickUp.
- Produced technical documentation, incorporated stakeholder feedback, and continuously improved software quality through problem-solving and adoption of development best practices.

Projects

AI Fitness Workout Optimisation System | *Final-Year Research – Python, Machine Learning, Computer Vision, Flutter*

- Developed a mobile application enabling daily workout tracking, where users upload exercise videos and an AI/ML-powered pose detection model evaluates exercise form correctness in real time.
- Integrated computer vision techniques to extract body keypoints and assess posture alignment, generating instant corrective feedback and personalised workout optimisation recommendations.
- Designed and built the Flutter-based mobile frontend and connected it to a Python backend hosting the trained ML models via RESTful APIs.
- Evaluated multiple pose estimation approaches and fine-tuned model performance to maximise accuracy across varied lighting conditions and body types.

DocBooker – Doctor Appointment Platform | *MongoDB, Express.js, React, Node.js (MERN)*

- Built a full-stack web application enabling patients to search for doctors and book medical appointments online, featuring a responsive UI, secure JWT-based user authentication, and a dynamic real-time scheduling system.
- Implemented three distinct role-based dashboards: a patient dashboard for managing bookings and medical history, a doctor dashboard for viewing schedules and patient records, and an admin dashboard for user and platform management.
- Developed RESTful APIs with Node.js and Express.js and used MongoDB for flexible data storage; integrated frontend and backend with Axios for seamless asynchronous communication.

PrimeResidency Application | *MERN Stack*

- Built a full-stack property booking web application allowing users to browse, search, filter, and book or purchase rooms and apartments.
- Developed two role-based dashboards: a user dashboard for managing bookings and profile details, and an admin dashboard for managing listings, users, and reservations.
- Implemented complete CRUD operations, image uploads, and responsive layouts optimised for desktop and mobile devices.

Online Logistics Management System | *MERN Stack*

- Developed a web application to manage end-to-end shipment and delivery workflows, including order creation, tracking, status updates, and delivery confirmation.

- Built a RESTful backend with Node.js, Express.js, and MongoDB, and implemented full CRUD operations for shipments, customers, and delivery agents.
- Designed a responsive frontend in React.js with dynamic tables, search and filter capabilities, and real-time status indicators.

Skill Connect | *UI/UX (Figma), Spring Boot, React, MongoDB*

- Designed comprehensive UI/UX wireframes and high-fidelity prototypes in Figma, then developed a skill-sharing social platform where users can post skills, like, comment, and follow other users.
- Built a Spring Boot REST API backend with MongoDB for data persistence, and developed the React.js frontend with component-based architecture and responsive design.
- Implemented user authentication, news feed functionality, and interactive social features including notifications and user profile management.

Crop Yield Prediction | *Academic Project – ML, Python*

- Researched and implemented four deep learning and tabular ML models – MLP, DCN, TabNet, and TabResNet – for crop yield prediction using Python, evaluating each model's accuracy, generalisation, and suitability for agricultural datasets.
- Performed data preprocessing, feature engineering, and hyperparameter tuning; analysed and visualised model performance metrics to identify the best-performing architecture.

Fit Club – Fitness Landing Page | *React, Tailwind CSS*

- Built a modern, fully responsive marketing site optimised for speed and cross-browser compatibility, with attention to clean layout and UI detail.

Online Grocery Ordering System | *Java, MySQL*

- Developed a Java and MySQL based ordering platform with product catalogue, cart, and order management features.

QA Testing – pixelssuite.com | *Manual & Automation Testing (Cypress)*

- Performed manual testing and built automation test scripts using Cypress for pixelssuite.com.

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL, Dart, PHP

AI / Machine Learning: Machine Learning, Deep Learning (MLP, DCN, TabNet, TabResNet), Computer Vision, Pose Estimation, AI Model Integration, Data Preprocessing

Frontend Development: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Figma (UI/UX Design)

Backend Development: Node.js, Express.js, Spring Boot, RESTful APIs

Mobile Development: Flutter, Firebase

Databases: MongoDB, MySQL, PostgreSQL

Testing & Tools: Manual Testing, Cypress Automation Testing, Git & GitHub, Postman, Agile/Scrum, Docker (Basic)

Personal Skills

Problem Solving, Analytical Thinking, Team Collaboration, Communication Skills, Leadership, Time Management, Adaptability, Quick Learner, Critical Thinking, Attention to Detail, Creativity & Innovation, Research Mindset

Certifications

AWS AI Practitioner Challenge – Udacity / Accenture

Prompt Design in Vertex AI – Google Cloud

Power Apps Canvas Apps – Microsoft Learn

Front-End Web Development – Centre for Open & Distance Learning (CODL), University of Moratuwa, Sri Lanka

Server-Side Web Programming – Centre for Open & Distance Learning (CODL), University of Moratuwa, Sri Lanka

Web Design for Beginners – Centre for Open & Distance Learning (CODL), University of Moratuwa, Sri Lanka

Python Programming – Centre for Open & Distance Learning (CODL), University of Moratuwa, Sri Lanka

Strengths

Artificial Intelligence & Machine Learning, Full-Stack MERN Development, Strong Problem-Solving & Analytical Skills, UI/UX Design & Responsive Development, Software Testing (Manual & Automation), REST API & Database Development, Quick Learner & Adaptable, Team Collaboration & Communication, Research & Innovation Mindset, Attention to Detail & Time Management

Languages

English: Proficient | **Tamil:** Fluent | **Sinhala:** Basic

References

Dr. Kapila Dissanayaka

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Miss. Fathima Fanoon

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